

1.Program

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [2, 4, 6, 8, 10]
```

```
plt.plot(x, y, marker='o')
```

```
plt.title("Simple Line Graph")
```

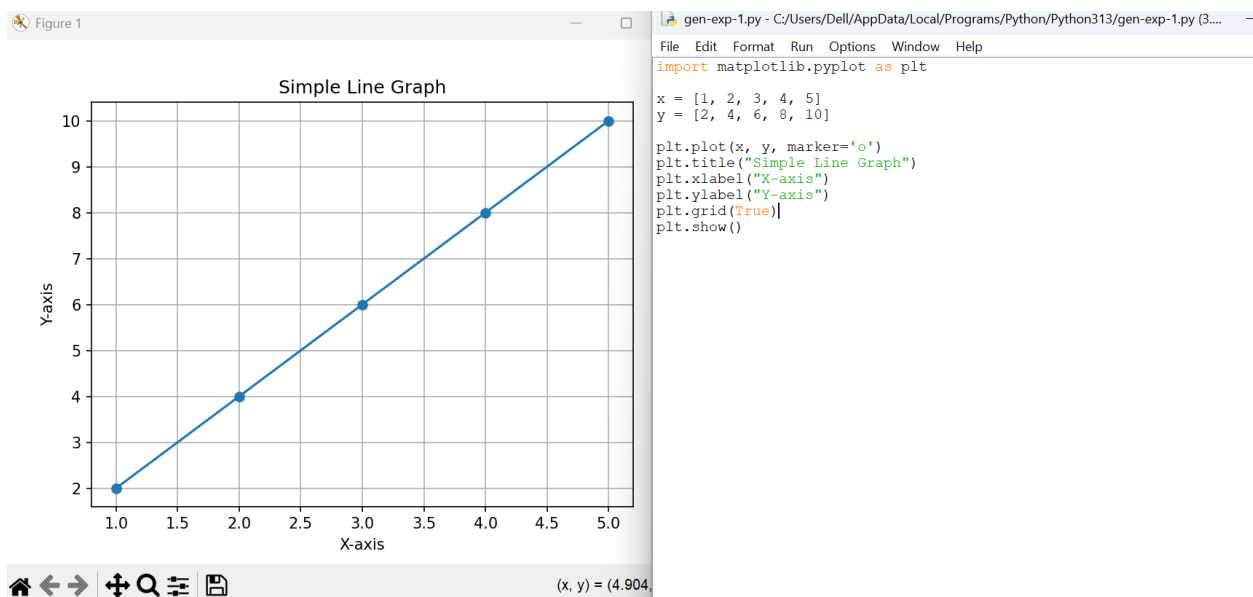
```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
plt.grid(True)
```

```
plt.show()
```

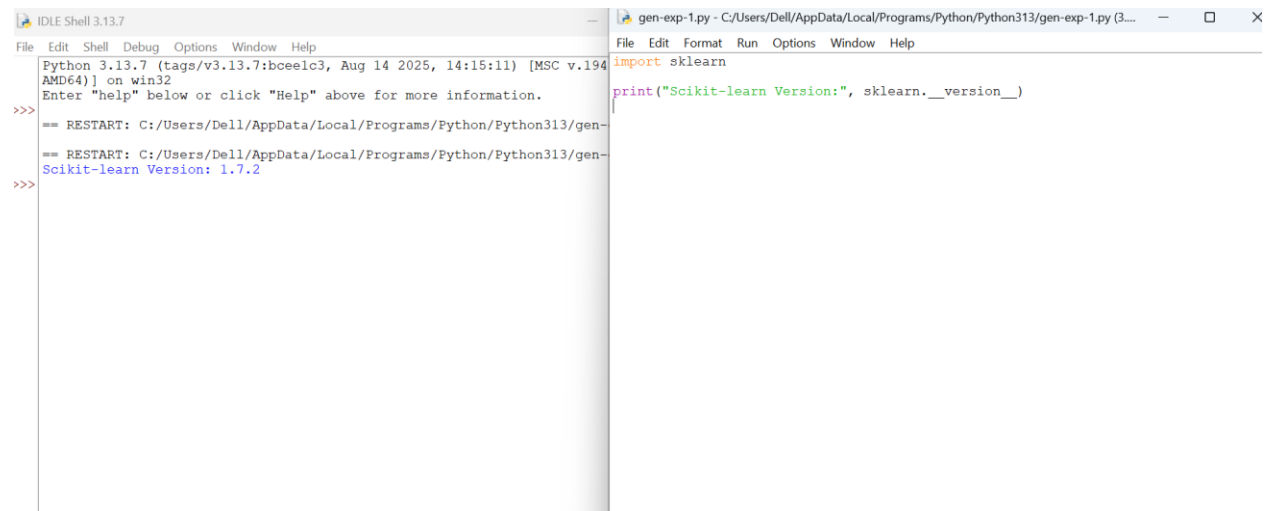
Output:



2.Program

```
import sklearn  
  
print("Scikit-learn Version:", sklearn.__version__)
```

Output:



The screenshot shows two windows. The left window is the IDLE Shell 3.13.7, which displays the output of the script: `Scikit-learn Version: 1.7.2`. The right window is a file editor showing the code for `gen-exp-1.py`, which contains the same code as shown in the previous block: `import sklearn` and `print("Scikit-learn Version:", sklearn.__version__)`.

3.Program

```
import numpy as np  
  
arr = np.arange(1, 21)  
  
print(arr)
```

Output:

```
Python 3.13.7 (tags/v3.13.7:bce1c3, Aug 14 2025, 14:15:11) [MSC v.1944 AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-
Scikit-learn Version: 1.7.2
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
>>>
```

```
gen-exp-1.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-exp-1.py (3...
File Edit Format Run Options Window Help
import numpy as np
arr = np.arange(1, 21)
print(arr)
```

4.Program

import pandas as pd

```
data = {
    "Name": ["Alice", "Bob", "Charlie", "David"],
    "Grade": [90, 85, 95, 88]
}
```

df = pd.DataFrame(data)

print(df)

Output:

```
Python 3.13.7 (tags/v3.13.7:bce1c3, Aug 14 2025, 14:15:11) [MSC v.1944 AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-
Scikit-learn Version: 1.7.2
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-
  Name  Grade
0  Alice    90
1    Bob    85
2  Charlie   95
3   David    88
>>>
```

```
gen-exp-1.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-exp-1.py (3...
File Edit Format Run Options Window Help
import pandas as pd
data = {
    "Name": ["Alice", "Bob", "Charlie", "David"],
    "Grade": [90, 85, 95, 88]
}
df = pd.DataFrame(data)
print(df)
```

5.Pogram

```
import numpy

import pandas

import matplotlib

import sklearn

import torch

import transformers

print("NumPy:", numpy.__version__)

print("Pandas:", pandas.__version__)

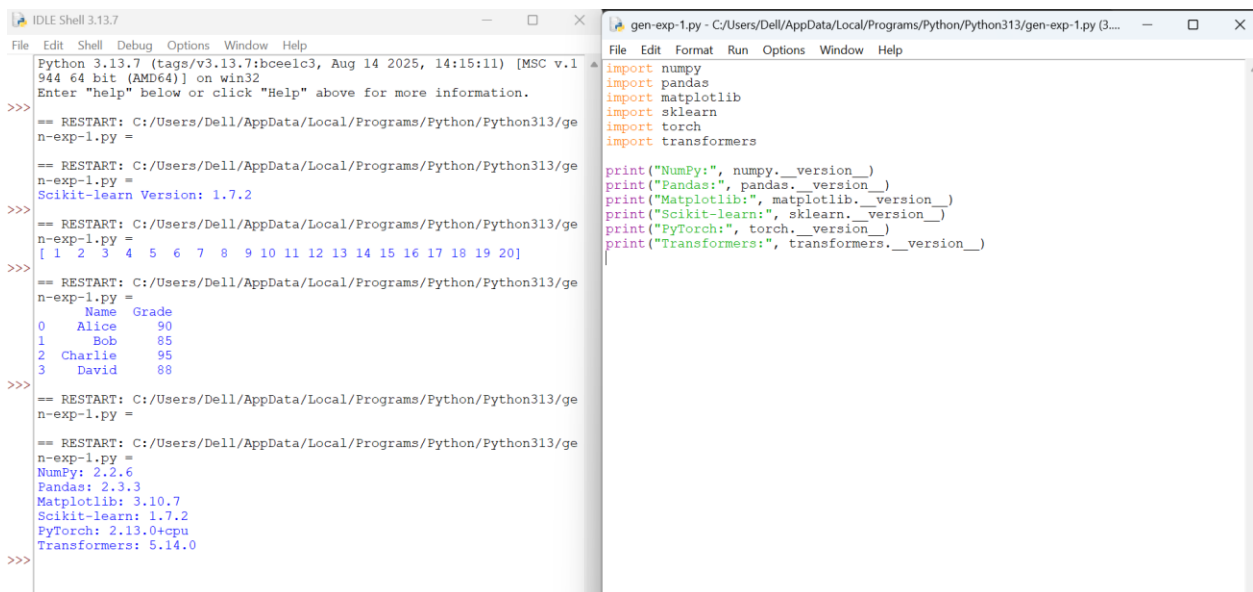
print("Matplotlib:", matplotlib.__version__)

print("Scikit-learn:", sklearn.__version__)

print("PyTorch:", torch.__version__)

print("Transformers:", transformers.__version__)
```

Output:



The image shows two side-by-side windows from a Python IDE. The left window, titled 'IDLE Shell 3.13.7', displays the output of a Python script. It shows several 'RESTART' messages and the execution of various modules, resulting in a list of numbers and a table of student grades. The right window, titled 'gen-exp-1.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/gen-exp-1.py (3...', shows the source code of the script, which imports several libraries and prints their versions.

```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1
944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/ge
n-exp-1.py =
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/ge
n-exp-1.py =
Skikit-learn Version: 1.7.2
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/ge
n-exp-1.py =
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/ge
n-exp-1.py =
      Name  Grade
0    Alice    90
1     Bob    85
2   Charlie    95
3    David    88
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/ge
n-exp-1.py =
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/ge
n-exp-1.py =
NumPy: 2.2.6
Pandas: 2.3.3
Matplotlib: 3.10.7
Scikit-learn: 1.7.2
PyTorch: 2.13.0+cpu
Transformers: 5.14.0
>>>
```

```
gen-exp-1.py
import numpy
import pandas
import matplotlib
import sklearn
import torch
import transformers

print("NumPy:", numpy.__version__)
print("Pandas:", pandas.__version__)
print("Matplotlib:", matplotlib.__version__)
print("Scikit-learn:", sklearn.__version__)
print("PyTorch:", torch.__version__)
print("Transformers:", transformers.__version__)
```

6.Program

```
import torch

a = torch.tensor([1,2,3])

b = torch.tensor([4,5,6])

print("Tensor A:", a)

print("Tensor B:", b)

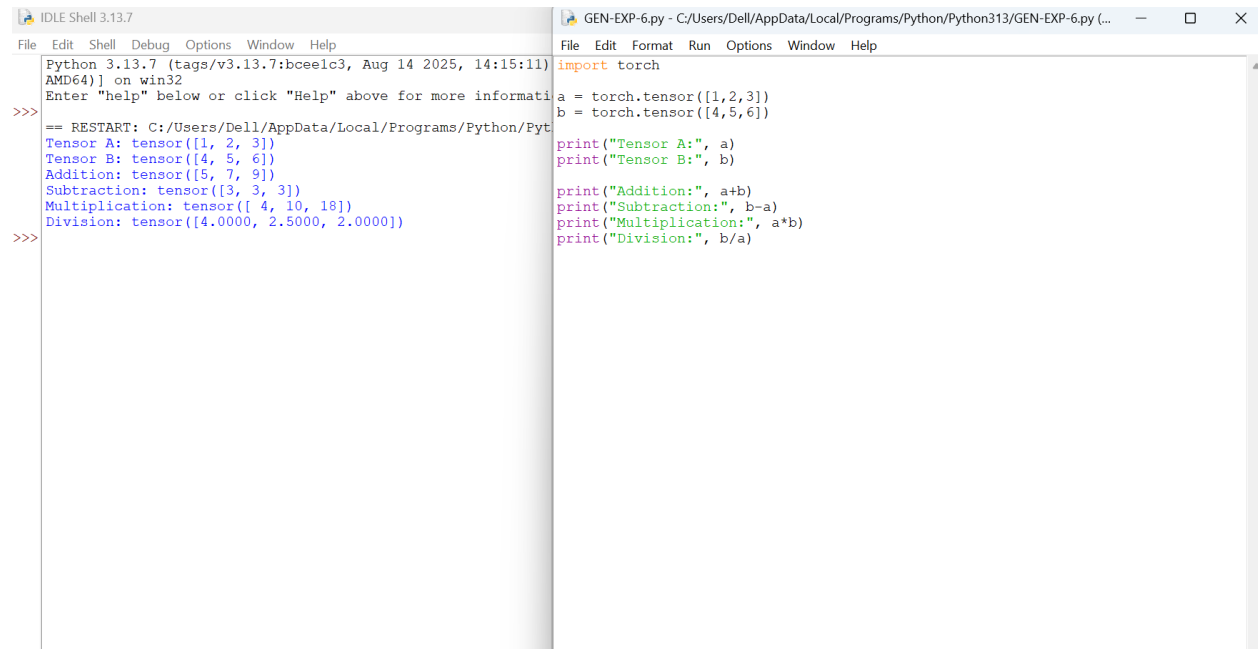
print("Addition:", a+b)

print("Subtraction:", b-a)

print("Multiplication:", a*b)

print("Division:", b/a)
```

Output:



```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11)
AMD64) on win32
Enter "help" below or click "Help" above for more informati
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Pyt
Tensor A: tensor([1, 2, 3])
Tensor B: tensor([4, 5, 6])
Addition: tensor([5, 7, 9])
Subtraction: tensor([3, 3, 3])
Multiplication: tensor([ 4, 10, 18])
Division: tensor([4.0000, 2.5000, 2.0000])
>>>
```

```
GEN-EXP-6.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EXP-6.py (...
File Edit Format Run Options Window Help
import torch
a = torch.tensor([1,2,3])
b = torch.tensor([4,5,6])
print("Tensor A:", a)
print("Tensor B:", b)
print("Addition:", a+b)
print("Subtraction:", b-a)
print("Multiplication:", a*b)
print("Division:", b/a)
```

7.Program

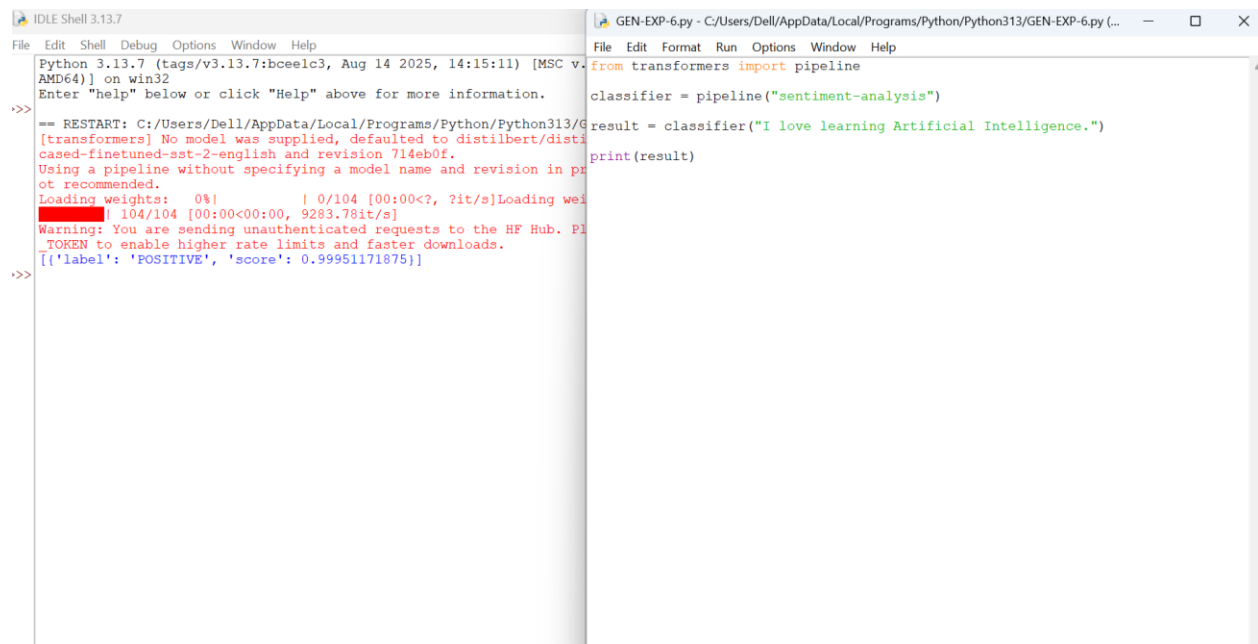
```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")

result = classifier("I love learning Artificial Intelligence.")

print(result)
```

Output:



```
IDLE Shell 3.13.7
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1916 AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/G
[transformers] No model was supplied, defaulted to distilbert/disti
cased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in pr
ot recommended.
Loading weights: 0%|          | 0/104 [00:00<?, ?it/s]Loading wei
██████████ 104/104 [00:00<00:00, 9283.78it/s]
Warning: You are sending unauthenticated requests to the HF Hub. Pl
_TOKEN to enable higher rate limits and faster downloads.
[{'label': 'POSITIVE', 'score': 0.99951171875}]
>>>
```

```
GEN-EXP-6.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EXP-6.py (...
File Edit Format Run Options Window Help
from transformers import pipeline

classifier = pipeline("sentiment-analysis")

result = classifier("I love learning Artificial Intelligence.")

print(result)
```

8.Program

```
from transformers import BertTokenizer

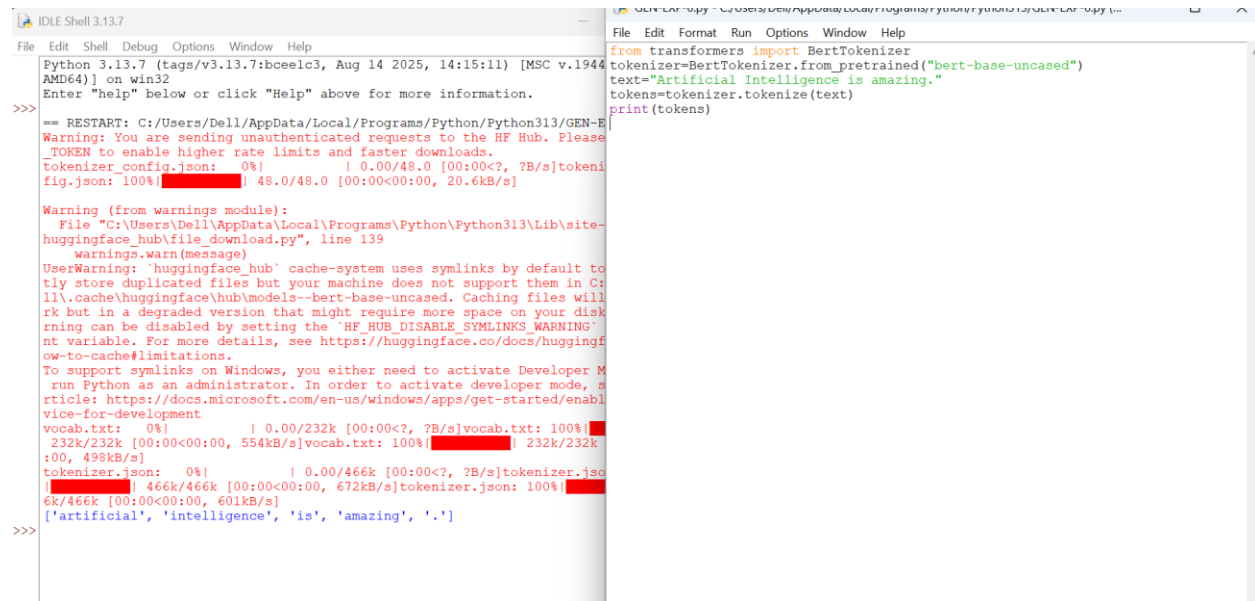
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

text = "Artificial Intelligence is amazing."

tokens = tokenizer.tokenize(text)

print(tokens)
```

Output:



```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944 AMD64] on win32
Enter "help" below or click "Help" above for more information.

>>>
== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-...
Warning: You are sending unauthenticated requests to the HF Hub. Please
TOKEN to enable higher rate limits and faster downloads.
tokenizer_config.json: 0%|          | 0.00/48.0 [00:00<, ?B/s]tokenizer
fig.json: 100%|██████████| 48.0/48.0 [00:00<00:00, 20.6kB/s]

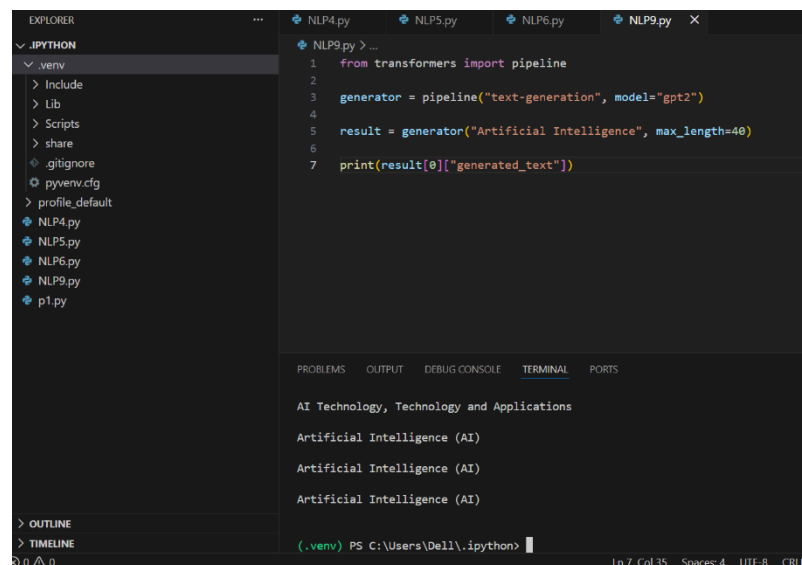
Warning (from warnings module):
  File "C:/Users/Dell/AppData/Local/Programs/Python/Python313/Lib/site-
huggingface_hub/file_download.py", line 139
    warnings.warn(message)
UserWarning: 'huggingface_hub' cache-system uses symlinks by default to
tly store duplicated files but your machine does not support them in C:
ll.cache\huggingface\hub\models--bert-base-uncased. Caching files will
rk but in a degraded version that might require more space on your disk
ring can be disabled by setting the 'HF_HUB_DISABLE_SYMLINKS_WARNING'
nt variable. For more details, see https://huggingface.co/docs/huggingf
ow-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer M
run Python as an administrator. In order to activate developer mode, s
rticle: https://docs.microsoft.com/en-us/windows/apps/get-started/enabl
vice-for-development
vocab.txt: 0%|          | 0.00/232k [00:00<, ?B/s]vocab.txt: 100%|███████
232k/232k [00:00<00:00, 554kB/s]vocab.txt: 100%|██████████| 232k/232k
:00, 498kB/s]
tokenizer.json: 0%|          | 0.00/466k [00:00<, ?B/s]tokenizer.js
██████████| 466k/466k [00:00<00:00, 672kB/s]tokenizer.json: 100%|███████
6k/466k [00:00<00:00, 601kB/s]
['artificial', 'intelligence', 'is', 'amazing', '.']

>>>
```

9.Program

```
from transformers import pipeline
generator = pipeline("text-generation", model="gpt2")
result = generator("Artificial Intelligence", max_length=40)
print(result[0]["generated_text"])
```

Output:



```
EXPLORER | JUPYTHON | NLP4.py | NLP5.py | NLP6.py | NLP9.py | X
├─ .venv
│  ├─ Include
│  ├─ Lib
│  ├─ Scripts
│  ├─ share
│  ├─ gitignore
│  └─ pyvenv.cfg
├─ profile.default
├─ NLP4.py
├─ NLP5.py
├─ NLP6.py
├─ NLP9.py
└─ p1.py

NLP9.py | 1 from transformers import pipeline
        2
        3 generator = pipeline("text-generation", model="gpt2")
        4
        5 result = generator("Artificial Intelligence", max_length=40)
        6
        7 print(result[0]["generated_text"])

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

AI Technology, Technology and Applications

Artificial Intelligence (AI)

Artificial Intelligence (AI)

Artificial Intelligence (AI)

(.venv) PS C:\Users\Dell\.ipython>
```

10.Program

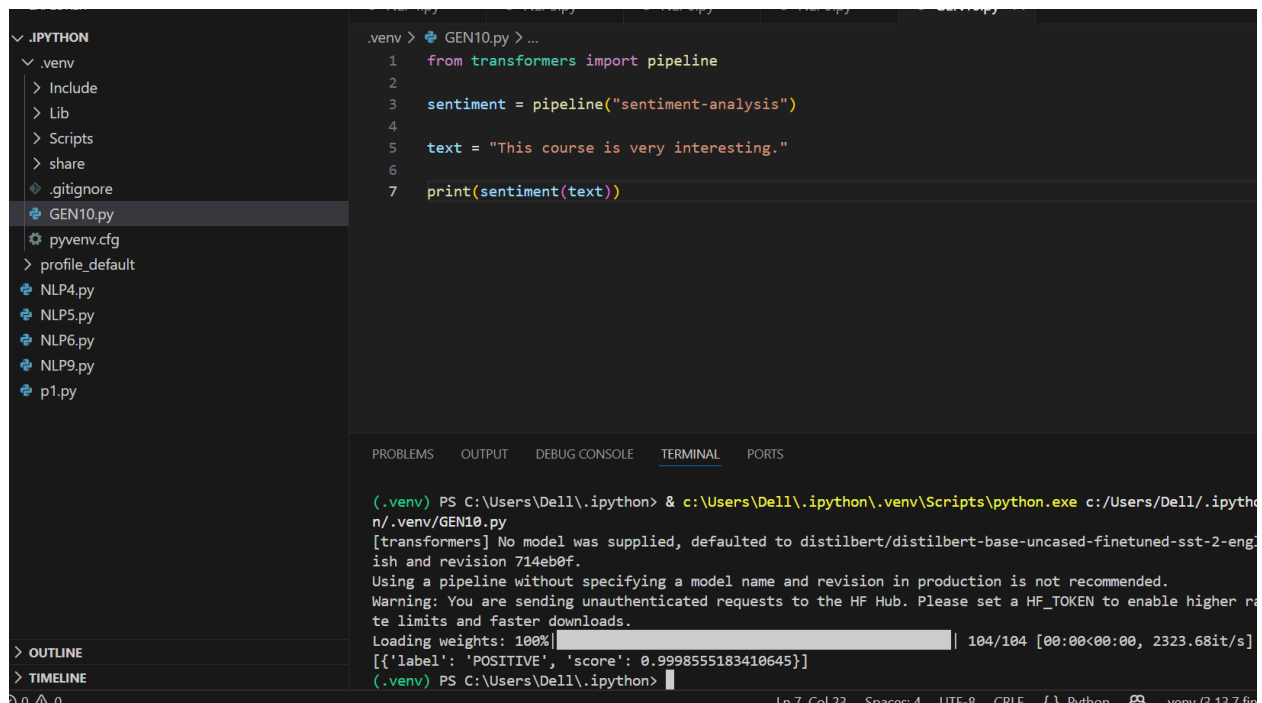
```
from transformers import pipeline

sentiment = pipeline("sentiment-analysis")

text = "This course is very interesting."

print(sentiment(text))
```

Output:



```
.venv > GEN10.py > ...
1 from transformers import pipeline
2
3 sentiment = pipeline("sentiment-analysis")
4
5 text = "This course is very interesting."
6
7 print(sentiment(text))
```

```
(.venv) PS C:\Users\Dell\.ipython> & c:\Users\Dell\.ipython\.venv\Scripts\python.exe c:/Users/Dell/.ipython/.venv/GEN10.py
[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 104/104 [00:00<00:00, 2323.68it/s]
[{'label': 'POSITIVE', 'score': 0.9998555183410645}]
(.venv) PS C:\Users\Dell\.ipython>
```

11.Program

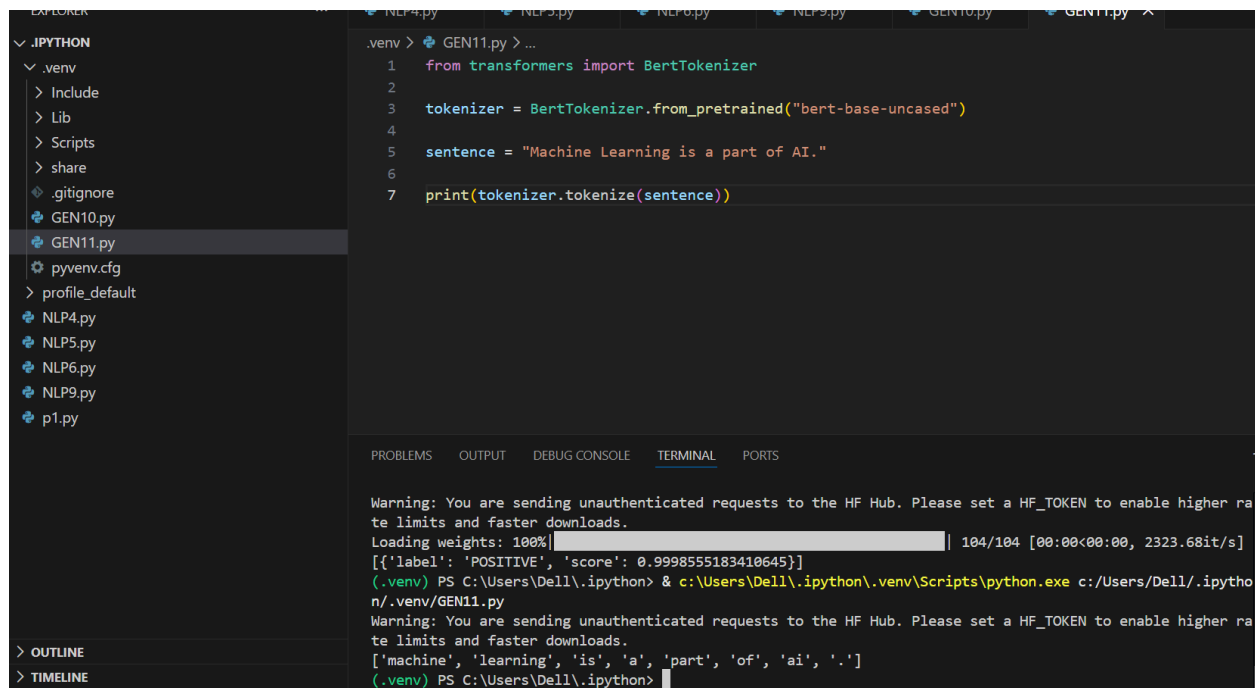
```
from transformers import BertTokenizer

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

sentence = "Machine Learning is a part of AI."

print(tokenizer.tokenize(sentence))
```


Output:



The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with a .venv directory and several Python files. The code editor shows the content of GEN11.py, which imports BertTokenizer from transformers and prints the tokenized output of a sentence. The terminal at the bottom shows the execution of the script, including a warning about unauthenticated requests to the Hugging Face Hub and the tokenized output of the sentence.

```
.venv > GEN11.py > ...
1  from transformers import BertTokenizer
2
3  tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
4
5  sentence = "Machine Learning is a part of AI."
6
7  print(tokenizer.tokenize(sentence))
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 104/104 [00:00<00:00, 2323.68it/s]
[{'label': 'POSITIVE', 'score': 0.9998555183410645}]
(.venv) PS C:\Users\Dell\.ipython> & c:\Users\Dell\.ipython\.venv\Scripts\python.exe c:/Users/Dell/.ipython/.venv/GEN11.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
['machine', 'learning', 'is', 'a', 'part', 'of', 'ai', '.']
(.venv) PS C:\Users\Dell\.ipython>

13.Program

```
from transformers import GPT2Tokenizer, GPT2LMHeadModel
```

```
import torch
```

```
tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
```

```
model = GPT2LMHeadModel.from_pretrained("gpt2")
```

```
input_text = "The future of AI"
```

```
input_ids = tokenizer.encode(input_text, return_tensors="pt")
```

```
output = model.generate(input_ids, max_length=50)
```

```
print(tokenizer.decode(output[0], skip_special_tokens=True))
```

```
IDLE Shell 3.13.7
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944 AMD64] on win32
Enter "help" below or click "Help" above for more information.
> = RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EXP
= RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EXP
Warning: You are sending unauthenticated requests to the HF Hub. Please
TOKEN to enable higher rate limits and faster downloads.
Loading weights:   0% | 0/148 [00:00<?, ?it/s] Loading weights:
███████ 148/148 [00:00<00:00, 3544.21it/s]
The future of AI is uncertain. The future of AI is uncertain.
The future of AI is uncertain. The future of AI is uncertain.
The future of AI is uncertain. The future of AI is uncertain.
The future
```

```
from transformers import BertTokenizer

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

text = "Natural Language Processing"

tokens = tokenizer.tokenize(text)

ids = tokenizer.convert_tokens_to_ids(tokens)

print("Tokens:", tokens)

print("Token IDs:", ids)
```

```
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bee51c3, Aug 14 2025, 14:15:11) [MSC v.1916
AMD64] on win32
Enter "help()" below or click "Help" above for more information.
>>
= RESTART: C:\Users\ DELL\AppData\Local\Programs\Python\Python313\GEN-EX
Tokens: ['natural', 'language', 'processing']
Token IDs: [3019, 2653, 6364]

File Edit Format Run Options Window Help
from transformers import BertTokenizer
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

text = "Natural Language Processing"
tokens = tokenizer.tokenize(text)
ids = tokenizer.convert_tokens_to_ids(tokens)

print("Tokens:", tokens)
print("Token IDs:", ids)
```

16.Program

```
from transformers import BertTokenizer, GPT2Tokenizer

bert = BertTokenizer.from_pretrained("bert-base-uncased")

gpt = GPT2Tokenizer.from_pretrained("gpt2")

text = "Natural Language Processing"

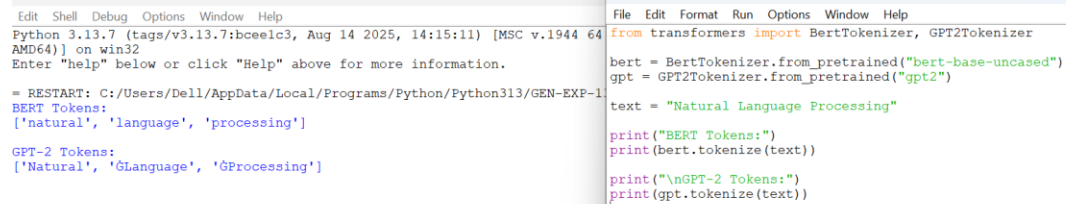
print("BERT Tokens:")

print(bert.tokenize(text))

print("\nGPT-2 Tokens:")

print(gpt.tokenize(text))
```

Output:



```
File Edit Format Run Options Window Help
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944 64
AMD64]) on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EXP-1
BERT Tokens:
['natural', 'language', 'processing']

GPT-2 Tokens:
['Natural', 'GLanguage', 'GProcessing']

File Edit Format Run Options Window Help
from transformers import BertTokenizer, GPT2Tokenizer

bert = BertTokenizer.from_pretrained("bert-base-uncased")
gpt = GPT2Tokenizer.from_pretrained("gpt2")

text = "Natural Language Processing"

print("BERT Tokens:")
print(bert.tokenize(text))

print("\nGPT-2 Tokens:")
print(gpt.tokenize(text))
```

17.Program

```
from transformers import BertTokenizer, BertModel

import torch

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

model = BertModel.from_pretrained("bert-base-uncased")

sentence = "Artificial Intelligence is transforming the world."

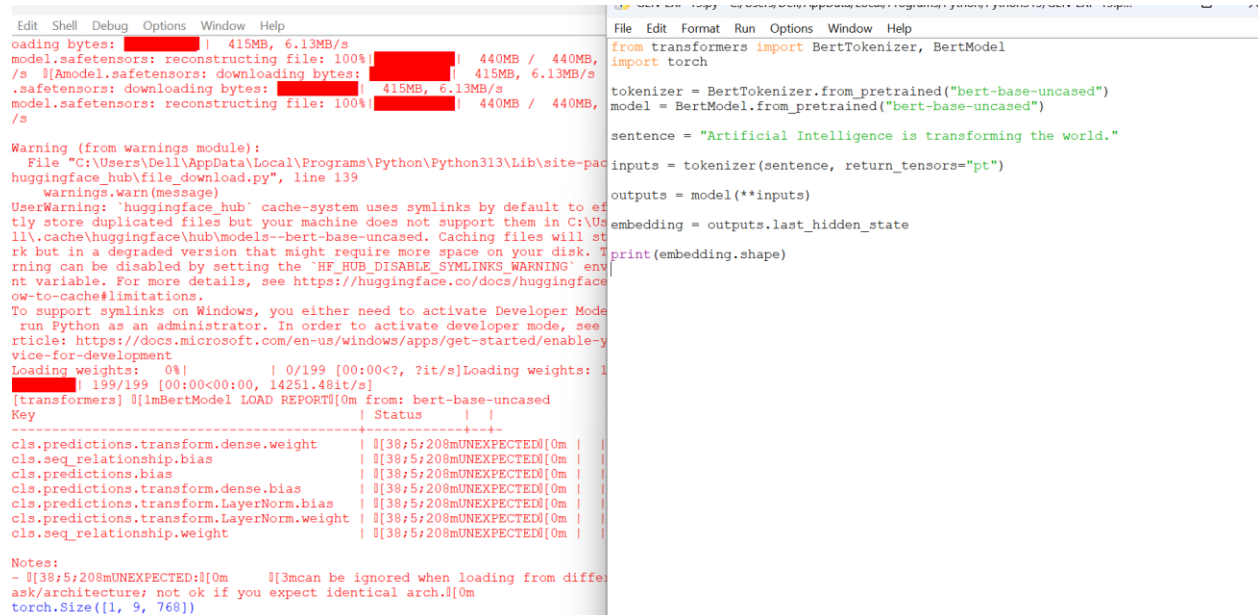
inputs = tokenizer(sentence, return_tensors="pt")

outputs = model(**inputs)

embedding = outputs.last_hidden_state

print(embedding.shape)
```

Output:



```
File Edit Shell Debug Options Window Help
loading bytes: 415MB, 6.13MB/s
model.safetensors: reconstructing file: 100% 440MB / 440MB
/s [Amodel.safetensors: downloading bytes: 415MB, 6.13MB/s
.safetensors: downloading bytes: 415MB, 6.13MB/s
model.safetensors: reconstructing file: 100% 440MB / 440MB
/s

Warning (from warnings module):
File "C:\Users\Dell\AppData\Local\Programs\Python\Python313\Lib\site-packages\huggingface_hub\file_download.py", line 139
warnings.warn(message)
UserWarning: 'huggingface_hub' cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\Dell\cache\huggingface\hub\models--bert-base-uncased. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the 'HF_HUB_DISABLE_SYMLINKS_WARNING' environment variable. For more details, see https://huggingface.co/docs/huggingface_hub-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or run Python as an administrator. In order to activate developer mode, see article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
Loading weights: 0% | 0/199 [00:00<?, ?it/s] Loading weights: 100% | 199/199 [00:00<00:00, 14251.48it/s]
[transformers] [lmBertModel LOAD REPORT][0m from: bert-base-uncased
Key | Status |
-----+-----
cls.predictions.transform.dense.weight | [38;5;208mUNEXPECTED[0m |
cls.seq_relationship.bias | [38;5;208mUNEXPECTED[0m |
cls.predictions.bias | [38;5;208mUNEXPECTED[0m |
cls.predictions.transform.dense.bias | [38;5;208mUNEXPECTED[0m |
cls.predictions.transform.LayerNorm.bias | [38;5;208mUNEXPECTED[0m |
cls.predictions.transform.LayerNorm.weight | [38;5;208mUNEXPECTED[0m |
cls.seq_relationship.weight | [38;5;208mUNEXPECTED[0m |

Notes:
- [38;5;208mUNEXPECTED[0m [3mcan be ignored when loading from different
ask/architecture; not ok if you expect identical arch.[0m
torch.Size([1, 9, 768])
```

18.Program

```
from transformers import BertTokenizer, BertModel

import torch

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

model = BertModel.from_pretrained("bert-base-uncased")

sent1 = "The cat is sitting on the mat."

sent2 = "A cat is sitting on a mat."

inputs1 = tokenizer(sent1, return_tensors="pt")

inputs2 = tokenizer(sent2, return_tensors="pt")

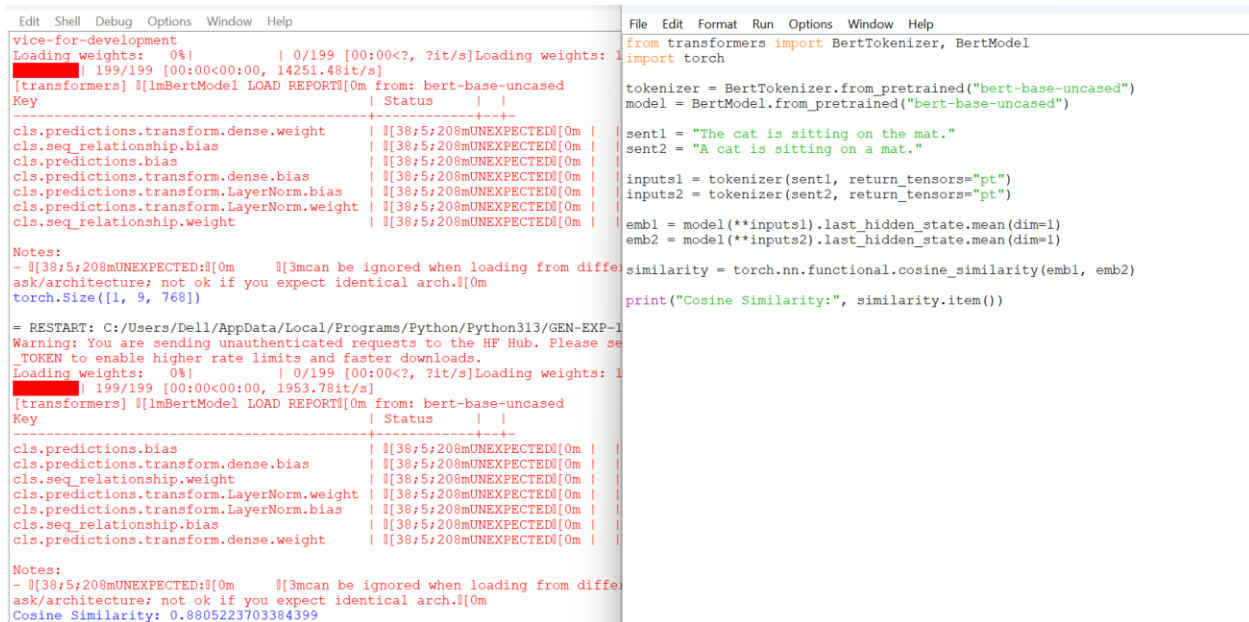
emb1 = model(**inputs1).last_hidden_state.mean(dim=1)

emb2 = model(**inputs2).last_hidden_state.mean(dim=1)

similarity = torch.nn.functional.cosine_similarity(emb1, emb2)

print("Cosine Similarity:", similarity.item())
```

Output:



```
File Edit Format Run Options Window Help
vice-for-development
Loading weights: 0% | 0/199 [00:00<?, ?it/s] Loading weights: 199/199 [00:00<00:00, 14251.48it/s]
[transformers] [lmBertModel LOAD REPORT][0m from: bert-base-uncased
Key | Status |
-----+-----+
cls.predictions.transform.dense.weight | [38;5;208mUNEXPECTED][0m |
cls.seq_relationship.bias | [38;5;208mUNEXPECTED][0m |
cls.predictions.bias | [38;5;208mUNEXPECTED][0m |
cls.predictions.transform.dense.bias | [38;5;208mUNEXPECTED][0m |
cls.predictions.transform.LayerNorm.bias | [38;5;208mUNEXPECTED][0m |
cls.predictions.transform.LayerNorm.weight | [38;5;208mUNEXPECTED][0m |
cls.seq_relationship.weight | [38;5;208mUNEXPECTED][0m |
Notes:
- [38;5;208mUNEXPECTED:[0m [3mcan be ignored when loading from differ
ask/architecture; not ok if you expect identical arch.[0m
torch.Size([1, 9, 768])
= RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EXP-1
Warning: You are sending unauthenticated requests to the HF Hub. Please se
TOKEN to enable higher rate limits and faster downloads.
Loading weights: 0% | 0/199 [00:00<?, ?it/s] Loading weights: 199/199 [00:00<00:00, 1953.78it/s]
[transformers] [lmBertModel LOAD REPORT][0m from: bert-base-uncased
Key | Status |
-----+-----+
cls.predictions.bias | [38;5;208mUNEXPECTED][0m |
cls.predictions.transform.dense.bias | [38;5;208mUNEXPECTED][0m |
cls.seq_relationship.weight | [38;5;208mUNEXPECTED][0m |
cls.predictions.transform.LayerNorm.weight | [38;5;208mUNEXPECTED][0m |
cls.predictions.transform.LayerNorm.bias | [38;5;208mUNEXPECTED][0m |
cls.seq_relationship.bias | [38;5;208mUNEXPECTED][0m |
cls.predictions.transform.dense.weight | [38;5;208mUNEXPECTED][0m |
Notes:
- [38;5;208mUNEXPECTED:[0m [3mcan be ignored when loading from differ
ask/architecture; not ok if you expect identical arch.[0m
Cosine Similarity: 0.8805223703384399
```

19.Program

```
from transformers import pipeline

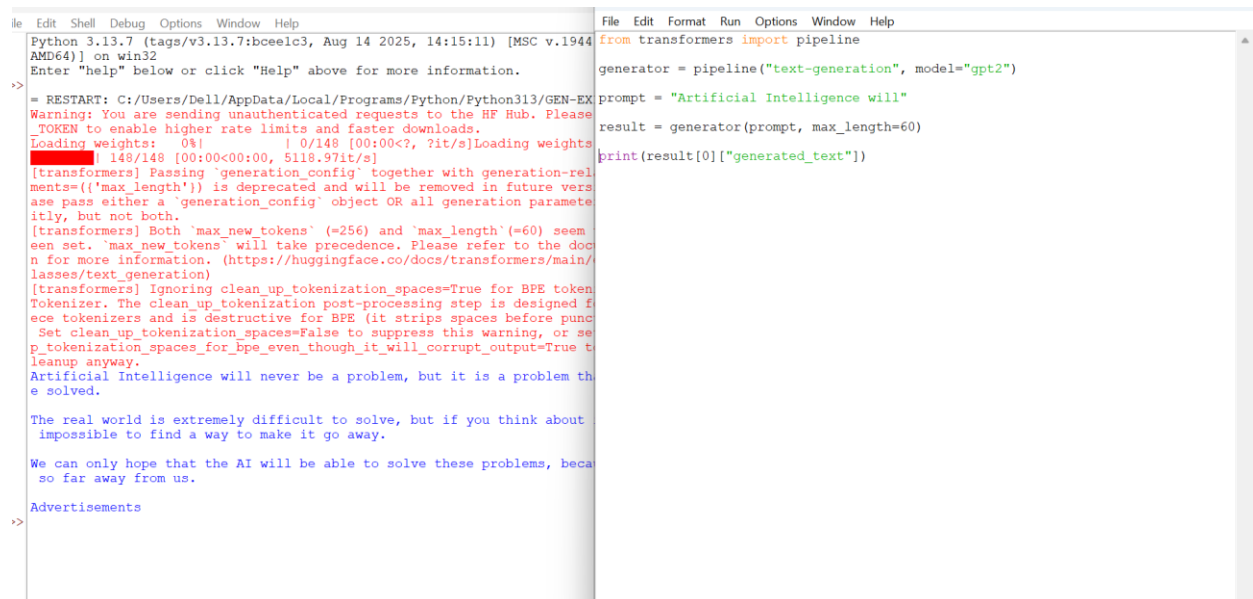
generator = pipeline("text-generation", model="gpt2")

prompt = "Artificial Intelligence will"

result = generator(prompt, max_length=60)

print(result[0]["generated_text"])
```

Output:



```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944 AMD64] on win32
Enter "help" below or click "Help" above for more information.
>
= RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EX
Warning: You are sending unauthenticated requests to the HF Hub. Please
_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 0% | 0/148 [00:00<?, ?it/s]Loading weights
| 148/148 [00:00<00:00, 5118.97it/s]
[transformers] Passing 'generation_config' together with generation-rel
ments=({'max_length'}) is deprecated and will be removed in future vers
ase pass either a 'generation_config' object OR all generation paramete
itly, but not both.
[transformers] Both 'max_new_tokens' (=256) and 'max_length' (=60) seem
een set. 'max_new_tokens' will take precedence. Please refer to the doc
n for more information. (https://huggingface.co/docs/transformers/main/
lasses/text_generation)
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE token
Tokenizer. The clean_up_tokenization post-processing step is designed f
ece tokenizers and is destructive for BPE (it strips spaces before punc
Set clean_up_tokenization_spaces=False to suppress this warning, or se
p_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True t
leanup anyway.
Artificial Intelligence will never be a problem, but it is a problem th
e solved.

The real world is extremely difficult to solve, but if you think about
impossible to find a way to make it go away.

We can only hope that the AI will be able to solve these problems, beca
so far away from us.

Advertisements
>
```

```
File Edit Format Run Options Window Help
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = "Artificial Intelligence will"

result = generator(prompt, max_length=60)

print(result[0]["generated_text"])
```

20.Program

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompts = [

    "Machine learning is",

    "Python programming",

    "Future of healthcare"

]

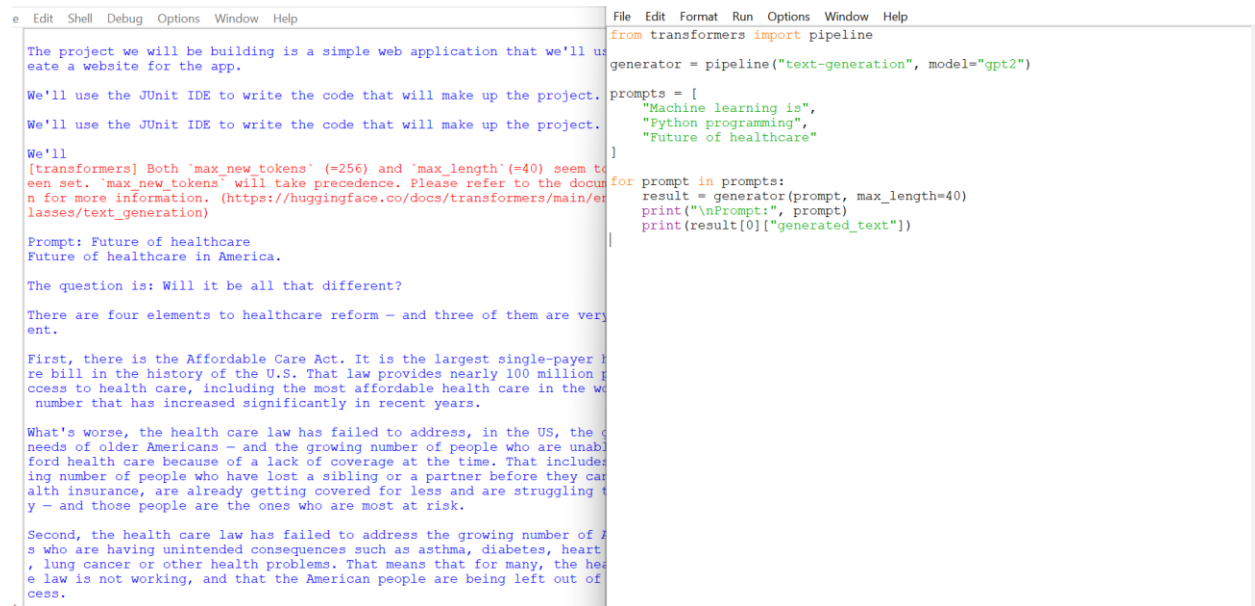
for prompt in prompts:

    result = generator(prompt, max_length=40)

    print("\nPrompt:", prompt)

    print(result[0]["generated_text"])
```

Output:



```
e Edit Shell Debug Options Window Help
The project we will be building is a simple web application that we'll use to create a website for the app.
We'll use the JUnit IDE to write the code that will make up the project.
We'll use the JUnit IDE to write the code that will make up the project.
We'll
[transformers] Both 'max_new_tokens' (~256) and 'max_length' (~40) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/advanced_text_generation)
Prompt: Future of healthcare
Future of healthcare in America.
The question is: Will it be all that different?
There are four elements to healthcare reform – and three of them are very important.
First, there is the Affordable Care Act. It is the largest single-payer health care bill in the history of the U.S. That law provides nearly 100 million people with access to health care, including the most affordable health care in the world. The number that has increased significantly in recent years.
What's worse, the health care law has failed to address, in the US, the needs of older Americans – and the growing number of people who are unable to afford health care because of a lack of coverage at the time. That includes a growing number of people who have lost a sibling or a partner before they can get health insurance, are already getting covered for less and are struggling to pay – and those people are the ones who are most at risk.
Second, the health care law has failed to address the growing number of people who are having unintended consequences such as asthma, diabetes, heart disease, lung cancer or other health problems. That means that for many, the health care law is not working, and that the American people are being left out of the health care system.
```

```
File Edit Format Run Options Window Help
from transformers import pipeline
generator = pipeline("text-generation", model="gpt2")
prompts = [
    "Machine learning is",
    "Python programming",
    "Future of healthcare"
]
for prompt in prompts:
    result = generator(prompt, max_length=40)
    print("\nPrompt:", prompt)
    print(result[0]["generated_text"])
```

12.Program

```
from transformers import BertTokenizer, BertModel

import torch

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

model = BertModel.from_pretrained("bert-base-uncased")

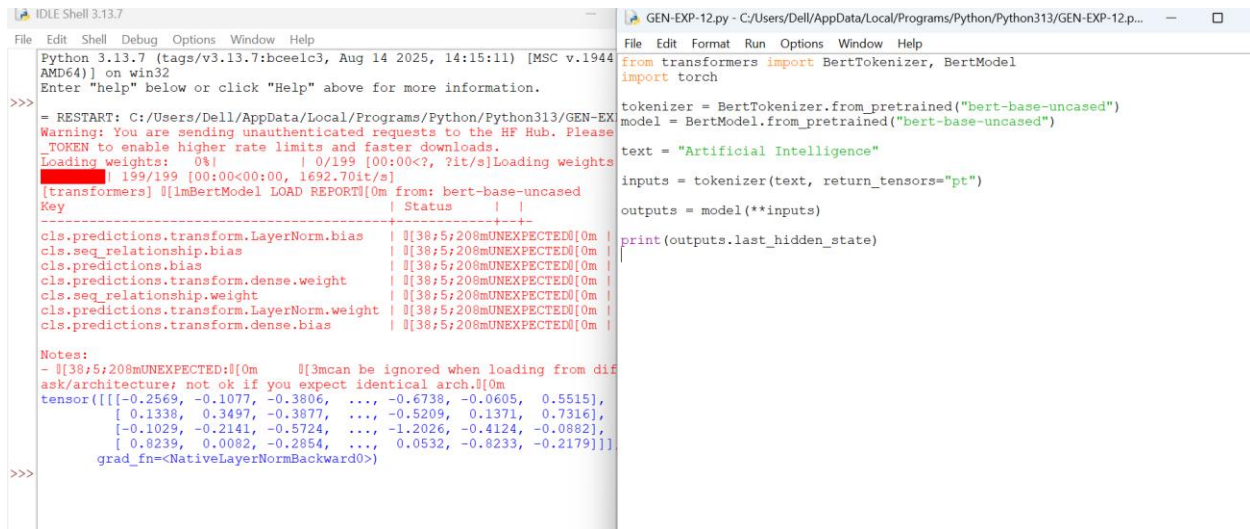
text = "Artificial Intelligence"

inputs = tokenizer(text, return_tensors="pt")

outputs = model(**inputs)

print(outputs.last_hidden_state)
```

Output:



```
Python 3.13.7 (tags/v3.13.7:bce1c3, Aug 14 2025, 14:15:11) [MSC v.1944 AMD64] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/GEN-EXP-12.py
Warning: You are sending unauthenticated requests to the HF Hub. Please
_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 0% | 0/199 [00:00<?, ?it/s]Loading weights
[199/199 [00:00<00:00, 1692.70it/s]
[transformers] [lmBertModel LOAD REPORT] from: bert-base-uncased
Key | Status |
-----+-----
cls.predictions.transform.LayerNorm.bias | [[38;5;208mUNEXPECTED[0m |
cls.seq_relationship.bias | [[38;5;208mUNEXPECTED[0m |
cls.predictions.bias | [[38;5;208mUNEXPECTED[0m |
cls.predictions.transform.dense.weight | [[38;5;208mUNEXPECTED[0m |
cls.seq_relationship.weight | [[38;5;208mUNEXPECTED[0m |
cls.predictions.transform.LayerNorm.weight | [[38;5;208mUNEXPECTED[0m |
cls.predictions.transform.dense.bias | [[38;5;208mUNEXPECTED[0m |

Notes:
- [[38;5;208mUNEXPECTED[0m [[3mcan be ignored when loading from dif
ask/architecture; not ok if you expect identical arch.[[0m
tensor([[[[-0.2569, -0.1077, -0.3806, ..., -0.6738, -0.0605, 0.5515],
[ 0.1338, 0.3497, -0.3877, ..., -0.5209, 0.1371, 0.7316],
[-0.1029, -0.2141, -0.5724, ..., -1.2026, -0.4124, -0.0882],
[ 0.8239, 0.0082, -0.2854, ..., 0.0532, -0.8233, -0.2179]]]]
grad_fn=<NativeLayerNormBackward0>

>>>
```


14.Program

```
from transformers import pipeline
```

```
qa = pipeline("question-answering")
```

```
context = """
```

Python is a programming language developed by Guido van Rossum.

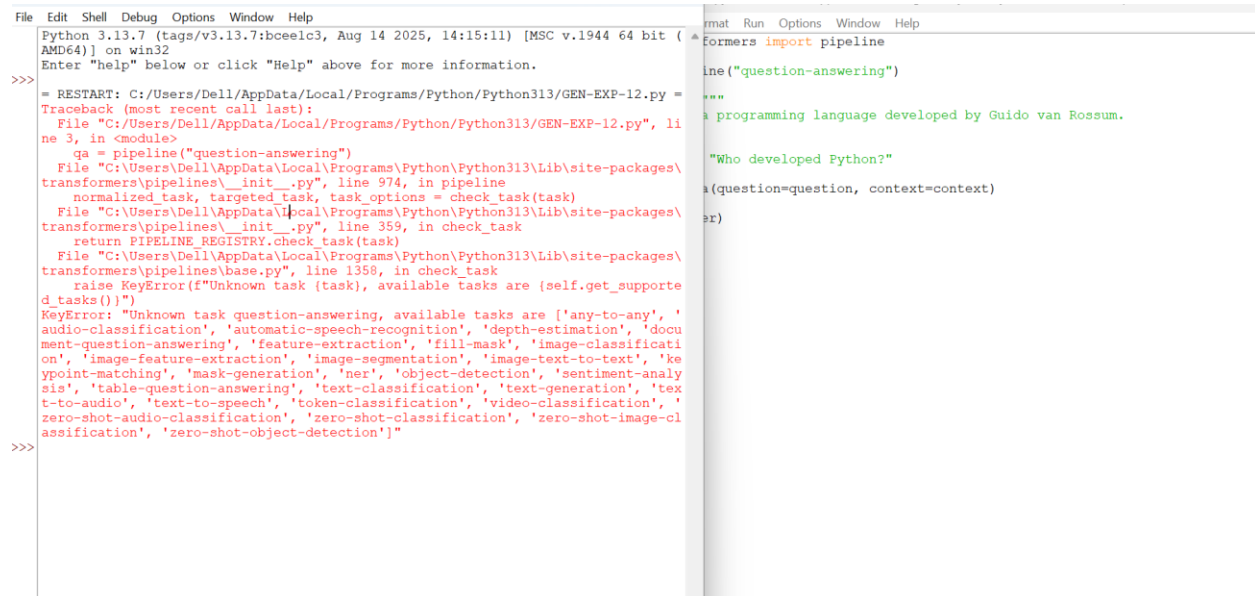
```
"""
```

```
question = "Who developed Python?"
```

```
answer = qa(question=question, context=context)
```

```
print(answer)
```

Output:



The screenshot shows a Python IDE with two panes. The left pane displays a traceback error for a `KeyError` with the message: "Unknown task question-answering, available tasks are ['any-to-any', 'audio-classification', 'automatic-speech-recognition', 'depth-estimation', 'document-question-answering', 'feature-extraction', 'fill-mask', 'image-classification', 'image-feature-extraction', 'image-segmentation', 'image-text-to-text', 'keypoint-matching', 'mask-generation', 'ner', 'object-detection', 'sentiment-analysis', 'table-question-answering', 'text-classification', 'text-generation', 'text-to-audio', 'text-to-speech', 'token-classification', 'video-classification', 'zero-shot-audio-classification', 'zero-shot-classification', 'zero-shot-image-classification', 'zero-shot-object-detection']". The right pane shows the code from the previous block, with the `qa` variable assigned to a `pipeline` object for the "question-answering" task, and the `question` and `context` variables defined.

